

FIT5147 Data Visualisation Project (DVP)

DVP Part 1: Design Presentation

DVP Part 2: Report and Code

In this project, you are asked to create an **interactive narrative visualisation** that communicates some of your findings from the *Data Exploration Project*. It is an **individual assignment** and is **worth 40%** of your total mark for FIT5147. The Data Visualisation Project (DVP) has two parts:

- **DVP Part 1: Design Presentation**, Weight: 3%. The mark is for the in-class presentation of your submitted design.
- **DVP Part 2: Report and Code**, Weight: 37%. Submission: Project report and files (source code and data).

Relevant Learning Outcomes

- LO3: Critically evaluate and interpret a data visualisation
- LO5: Choose an appropriate visualisation
- LO6: Implement interactive data visualisations using **R (Shiny)** or **JavaScript (D3)**.

Generative AI tools can be used to help with the written and code content of this assessment task, but this must be clearly declared in your report. See the Originality section of this document for further details.

Overview of the Assessment Tasks

Scope	Task
DVP Part 1: Design Presentation (3%)	1. Identify which findings from your Data Exploration Project you wish to communicate. You can be selective, and you do not need to share everything you have found. The visualisations and accompanying narration should reflect the answers to (one or more of) the questions in your Project Proposal. 2. Clearly define your intended audience. The audience might be the elderly, young children, your classmates, the general public, politicians or whoever you like, although the choice should make sense for the data and topic of choice. Your subsequent interactive narrative visualisation submission should be designed specifically for the intended audience. 3. Design an interactive narrative visualisation using the Five Design Sheet (FDS) methodology. 4. Prepare a short presentation based on your FDS - one sheet per slide. 5. Submit the five design sheet slides for your presentation at the end of Week 10. 6. Present your presentation to your Applied Session in Week 11 or 12.
DVP Part 2: Report and	7. Review and improve your FDS sheet designs in response to the feedback from the presentations.

Code (37%)	8. Implement your visualisation using either R (Shiny) or JavaScript (D3). The use of other visualisation libraries and packages is subject to approval by the teaching team (see the section “Notes on Implementation”). Note that <u>you are not allowed to use R Markdown</u>. 9. Write a report with all the relevant sections and add your FDS sheets into the appendix, export it to PDF. 10. Submit your report and your source code and datafiles (see the section “How to Submit”) at the start of the exam period.
------------	---

DVP Part 1: Design Presentation (3%)

The presentation is an opportunity to gain feedback on your designs from the teaching staff and peers. Prepare a **three minute** presentation based on your five design sheets. Your presentation should consist of only **6 slides** covering:

1. An **Introduction Slide**: Your name, project title, aims and motivation (one slide)
2. Each of your **Five Design Sheets** (i.e., one sheet per slide).

The Five Design Sheet methodology (FDS) must be completely followed. All components should be legible and readable. For your sheets, you may use pen and paper, and then digitise them (at high quality to ensure readability), or use digital pen technology. If you do import any images from external sources to your sheets, you must reference the source appropriately in the relevant design sheet.

Important: The slides you present in person in your Applied session in Week 11 or 12 must match those submitted on Moodle.

DVP Part 2: Report and Code (37%)

Part 2 examines your description and justification of your design process and final implemented interactive narrative visualisation, demonstrating the technical and theoretical skills you have learnt in the unit. The written report is a fundamental component of this work and is graded in conjunction with your code, the resulting interactive narrative visualisation it creates and your completed (and perhaps improved) five design sheets, submitted in the appendix.

Report Requirements

Write a **report of up to 15 pages** (excluding the cover page, table of contents, bibliography, and appendix, which together have a limit of 10 pages) that consists of the following:

A cover page

Containing a descriptive **project title** of your narrative visualisation and **your identity**: including your full name, student ID, Applied session number, and Teaching Associate’s name.

Section 1: Introduction [Approx. ½-1 page]

A precise and succinct description of what key findings and messages you wanted your narrative visualisation to convey, and who your **intended audience** is, including their needs, background and expectations.

Section 2: Design Process [Approx. 3-5 pages]

A description and justification of your narrative visualisation design process. This should briefly refer to each of your five design sheets (you must provide each of your 5 design sheets in the Appendix), and **justify your design choices based on the theoretical content of the unit** (throughout Weeks 1-12). These may include but are not limited to:

- choice of visualisation type,
- choice of visual variable(s),
- choice of typography,
- reasons for a particular colour palette,
- decisions for your layout structure,
- choice of genre and/or narrative style,
- choice of interaction techniques,
- identifying consistency in your design and interaction,
- aspects of Munzner's what-why-how framework,
- reflection on Tufte's design principles,
- reference to Gestalt laws,
- and aspects of the human visual system,

It is important that this section is not simply a description of which visual elements you chose to include in the different sheets, but must justify your ideas and the process that led you to your final design choices. Be sure to cite suitable references. All visualisations in your design process should be relevant to your data and there should be a clear indication of how your choice of audience has influenced your design process. Any changes you have made to the design submitted as DVP Part 1 should also be explained and justified here.

Section 3: Implementation

3.1 Technical Implementation [Approx. ½ - 1 page]

This section contains a high-level description of your implementation, including libraries used, references to external code sources such as templates, and reasons for any differences between your final design (the Sheet 5 realisation that was justified and explained in the Design Process section) and the implemented design (described in 3.3), if applicable. This should include any additional data management for visualisation purposes. You should briefly explain the reasons why your DVP project was challenging (e.g., extensive wrangling was required, advanced use of D3, complex designs etc). You are not required to explain the code in detail. References to any external templates or code sources.

If additional exploration occurred or new data has been added (with teaching staff permission) since your Data Exploration Project then explain this in this section. In particular what additional wrangling, management or data exploration you have done and why this was necessary.

3.2. Interactive Narrative Visualisation Implementation [Approx. 2-5 pages]

Document and describe your final implemented submission, including screenshots of your final implementation (see Figure notes below). This section should clearly explain the final implementation and how its narrative and interaction portrays the data insights to the defined audience.

3.3. Using the Implementation [Approx. ½ - 1 page]

This section contains instructions for how to run, view and use your interactive narrative visualisation. This should emphasise any parts of your visualisation that may be easily missed by a reader (e.g., some interaction you have implemented that might not be immediately visible).

Section 4: Conclusion and Reflection [Approx. ½ -1 page]

Summarise your findings and what you have achieved with your narrative visualisation. Reflect on what you have learnt in this project, including what in hindsight you might have

done differently to improve the result with respect to the visualisation design process and final implementation and any future work that you would like to do.

Appendix (max 10 pages including the 5 design sheets):

This must include your **five design sheets**. Make sure you provide clear images and any handwriting is readable. There are five additional pages for extra context if you need, however, please note these five pages are not part of the grading process and no extra grades can be gained from what is in your appendix. This is purely supplementary.

Bibliography:

Appropriate **references** of all resources that have influenced your work in IEEE or APA style (refer to the [Monash Uni library's guide](#) page for Monash citing and reference style). This should include any code templates, design influences and references to design theory, as well as any sources that have influenced any data insights. Any references named in this list are expected to be cited within your report or code.

Marking Criteria

DVP Part 1: Design Presentation [3%]

- **Quality of oral presentation** (confidence, speed, voice) **and quality of slides** (legibility, design, images) [1%].
- **Logical structure of the presentation** [1%]
- **Choice of design content**, including completeness, appropriate level, discussion of design, and implementation alternatives [1%].

The teaching staff may stop your presentation if you go overtime. This may result in the loss of marks and reduce the amount of feedback you get on your design.

DVP Part 2: Report and Code [37%]

When grading your submission, **all components** (i.e., the quality of your narrative visualisation design, technical implementation, and the written report) are taken into account:

Criteria	Below 50%	Pass (50%+)	Credit (60%+)	HD (80%+)
Design Process (9%): Quality, Description and Justification	Inappropriate or incomplete design process (based on Section 2 of the report and FDS in appendix). 0 if no demonstration of any design process is provided.	Appropriate design process described and satisfactory use of FDS methodology, with adequate choices and minimal justifications.	Good design process with correct use of FDS methodology and consideration of defined audience. Some evidence of connection between design choice reasoning and the taught material. Good choice of (in report and in FDS) figures, descriptions and justifications to demonstrate a clear and considerable brainstorming (Sheet 1) with a good breadth of ideas ; three complete and distinctive alternative full interface designs (Sheet 2,3,4) and the final realisation (Sheet 5). Sheet 5 is a refinement or combined version of Sheets 2-4. Clear evidence of evolution of design ideas through the FDS process. The design process clearly relates to the proposed project topic, data sources and exploration insights. If designs are updated based on feedback from DVP Part 1 (presentation), changes are documented and explained. Designs are adequately justified, with narrative and interaction components considered, particularly in the final realisation (Sheet 5).	Excellence throughout the design process with a wide variety of appropriate and creative design ideas demonstrated that suit the data, topic, narrative and audience. Clear evidence of significant considerations of the design choices with justifications and reasonings from the taught material (unit readings and literature) grounded by cited references. High-quality FDS with rigorous application of the methodology. Very clear and coherent progression from brainstorming (Sheet 1), through alternative designs (Sheets 2–4), to a detailed and implementable final design (Sheet 5). Sheets 2-5 include layout, narrative and interaction considerations and the final realisation is relevant, fully justified and well described. Reflection on the full design process is provided.
Final Implemented Design: Interactive Narrative Visualisation (13%)	Inappropriate final data visualisations implemented. 0 if implemented interactive narrative visualisation is not accessible / visible.	Adequate and mostly appropriate visualisations, with limited but suitable interactions and narrative. Implementation relates to some of the layout and elements on FDS Sheet 5.	Good choice of a variety of visualisations, with suitable interactions and narrative for the defined audience, which are demonstrated, documented and described. Appropriate attention to visualisation details such as clear labeling of axes, suitable legends, appropriate and clean visualisation layout(s), use of visual hierarchy, layout and typography. Appropriate and consistent use of colour (hue, value or saturation) and/or other visual variables. Narrative provides suitable signposting and messaging to the reader.	Exceptional quality of interactive narrative visualisation with meticulous attention to detail. The visualisations, interaction and narrative are well documented, justified and clearly explained with reference to the unit materials. The relevance and connection to FDS Sheet 5 is clearly indicated. Visualisation choices, interactions, visual variables and layout decisions including page navigation are clearly purposeful and support the narrative and audience needs. Interactions are intuitive and well-designed to enhance the user experience. Multiple views of the data are effectively linked. The narrative provides strong guidance

			Appropriate information about the data source(s) provided. Implementation relates to the layout and elements on FDS Sheet 5.	and signposting, with text and visual elements working together seamlessly. Visual content enriches the narrative, which is clear and easy to follow for the defined audience and fully conveyed through the implementation without reliance on the report. Reflection on the final implemented design is provided.
Technical Implementation and Data Handling (9%)	Inappropriate packages used. Troubleshooting needed to run code*. 0 if not programmed using RShiny or d3 0 if no code is provided. <i>* If we can not fix your code easily, we will be in touch via email to respond within 48 hours. Please keep aware of this and check for emails during the marking period.</i>	Suitable code that runs adequately the first time, or with very minor editing. Comments and structure are appropriate.	Good implementation with clear and suitable code. Suitable and demonstrated management of data fit for purpose. Code contains meaningful comments and is formatted and designed in such a way that it is easy to read and legible. Good description in 'Technical implementation' section including any changes from your realisation in FDS Sheet 5 for technical reasons. This should also include description and justification for any further exploration, improvements or additional data added (with permission). Reasonable guidance of how to run and use the implementation in the 'Using the Implementation' section.	High quality implementation with quality robust code and management of data (multiple, very large, or complex) for improved visualisation implementation. Sophisticated coding with RShiny or d3 demonstrated. Very clear description and guidance for 'Technical Implementation' and 'Using the Implementation'. Reflection on implementation and its complexity is provided.
Quality of Report: Continuity, Quality, Completeness and Thoroughness (6%)	Unsuitable report. Page limit exceeded. 0 if no report is provided.	Adequate report with logical structure with all the expected sections. Evidence of some continuity of proposed project topic.	Good quality of writing with minimal errors. Good grammar and spelling. Figures are clear and legible with readable font and font size, clear titles, legends and labelled axes in written English with no special characters such as underscores. Adequate referencing of figures, tables and literature. Suitable figure and table captions. Clear continuity of the project evident throughout: narrative visualisation presents data insights from questions proposed in the Project Proposal and discovered during the Data Exploration Project.	Professional complete report with excellence and fluidity of writing combined with good use of high quality figures/images. Clear and reflective conclusion summarising achievements and learning, with thoughtful consideration of areas for improvement, especially for visualisations. Attentive and correct in-text referencing for tables, figures and literature citations using an appropriate referencing style (APA7 or IEEE).

Some notes for success:

Design Process:

Your design must follow the Five Design Sheet process and provide all the required information according to the FDS template. The designs for Sheets 2-4 must be distinct from each other. The final design on Sheet 5 is expected to be a refinement of one of those sheets (or a combination of components from each).

Each design for Sheets 2-4 is expected to complete the narrative independently of the other designs. No design sheet should just respond to one of the questions or findings narrated by the visualisation. No design sheet should consist of a single visualisation technique, e.g., one graph. Every design should resemble a complete solution with a clear layout that follows a particular narration style.

In **Sheet 5** (and your final implementation) your final design's layout should include a location to provide information about the data used in this work and a link to the original data source. The design must also indicate how the interface will provide guidance to the audience on how they should use the interactive narrative visualisation.

Implemented Final Design:

The implemented visualisation must be a narrative. Elements of the design must tell the data story, using text and visualisation techniques to narrate how the data and the findings of the exploration process enable questions about the topic of interest to be answered.

Your implemented narrative visualisation should be **based on the result of your FDS process**. It does not need to follow it exactly. However, it should resemble the final design in Sheet 5. Small changes to your final design are allowed (e.g., layout, visualisation choices, navigation method, colour), but any such differences between your design and how it was implemented must be explained **and justified** in the *Implementation* section of your report (see rubric).

Technical Implementation:

As a rule of thumb, all *visualisation* packages and libraries that are included in this unit are allowed for your implementation. **This includes, but is not limited to:**

- For R Shiny: *ggplot2*, *ggmap*, *ggraph*, *Leaflet*, *Plotly*, *igraph*, *wordcloud*, etc.
- For D3: *D3* itself, *Leaflet*, *MapBox*, etc. **Libraries which act as high-level wrappers for D3 are NOT allowed (e.g., *C3.js*, *dimple*).**

If you are unsure if a particular visualisation package or library is allowed, please ask on the Ed discussion board or discuss with your Teaching team.

Tools or packages used for data wrangling, data cleaning, Shiny theming, HTML5 templating, CSS styling, etc. are not subject to these rules and can be used freely (i.e., for anything other than the visualisations themselves). However, you should not use server-side code, like *Django* or *node.js*, when implementing your design.

Any data used for your DVP must be read from the files submitted with your code.

For performance reasons, it is recommended that you **pre-format all of your data files** before loading them into R Shiny or D3. In other words, all data wrangling and cleaning steps (if any) should be performed outside of your narrative visualisation code. You are not required to include the code for data wrangling and cleaning as part of your submission. However, if you have done considerable work since your Data Exploration Project, then you should describe these steps in Section 3.1 Technical Implementation. Furthermore, if your data supplied is not the version post wrangling, the wrangling code will need to be done in R or Javascript and be included in the implementation code to ensure the teaching team can run the code from the submitted data files.

Be sure to **check that your code runs correctly**. If possible, check if your code works on other computers and operating systems. If you do not have access to another computer you can try checking via the Monash [MoVE platform](#).

If your code requires some steps for it to run, then be sure to make these very clear in *readme* notes for your marker and describe this in Section 3.3 *Using the Implementation* section of your report. Your code must run on your marker's computer on the first attempt for us to be able to mark your submission. Heavy deductions for code troubleshooting in Technical Implementation apply. If after some troubleshooting your marker is still unable to get the code to run, further deductions will occur as we will not be able to fully mark your final implemented design.

Report:

Your report should contain only high-quality images of your narrative visualisation and five design sheets, indicating interaction when relevant. Where possible, avoid using a single screenshot of the entire page since the resolution might be low; instead, crop and explain individual sections of the page. It is also recommended that you export your PDF using a local word processor (e.g., Microsoft Word), as exporting your document as a PDF directly from Google Docs will result in low-quality images. Make sure you can read and understand the PDF document and its images at A4 size without requiring further enlargement.

Originality

As this is academic work, it must be original and must clearly indicate what elements were your work and what are based on someone-else's work. If you are including facts, data, opinions or any other written or graphical information from another source, you must cite the source and reference the bibliographic details for the source, using the APA or IEEE style guide. Otherwise this may be considered **plagiarism**. Direct quotations or reproductions must also adhere to the appropriate APA or IEEE style. **Be sure to acknowledge sources that influence your code through your code comments and references in your bibliography. Do not copy complete designs from any external sources.**

For the same reason, you must **declare any software or third party service** you use to write or code your submitted work, including the use of **Generative AI tools**, such as ChatGPT, Grammarly and Google Translate. You must declare the tool(s) and prompt(s) you used at the end of your report (this is not included in the page limit). It is highly recommended that you thoroughly check and validate any generated material that you include or use as a basis of any of your assignment. Remember that AI-generated material can be biased, simplified or false and is often not high enough quality for this assessment.

If you are retaking this unit from a previous semester, please ensure you choose a completely new design and new visualisation code. The text, design and code cannot have been used in any other

unit. Likewise, you cannot reuse any code or written content that you have used in any previous assessment tasks for any units. The only self-plagiarism that is allowed is the questions you set in your Project Proposal this semester and reusing some R code from your Data Exploration Project this semester (if you wish). **Any other written content from your Proposal or DEP Part 2 may not be reused. It must be rewritten for your DVP.**

If concerns arise regarding the originality of your work – whether due to plagiarism, collusion, contract cheating, the use of unapproved software or the use of Generative AI tools without declaration – your academic integrity will be reviewed. Confirmed breaches of academic integrity may result in penalties affecting your assignment mark, this unit, or even your enrolment. If you need further information on academic integrity at Monash, we recommend you review the [academic integrity module](#).

Submission and Due Dates

NOTE: All submission times are in Melbourne, Australia local time.

DVP Part 1: Design Presentation

1. **Prepare a PDF file** containing all five of your design sheets.
2. Name the file *StudentName_StudentID_Presentation.pdf*
3. Submit it via Moodle under *Assessments* → *DVP Part 1: Design (3%)*.

Submit your **presentation slides** to Moodle, due early **Week 11 (see Moodle for date and time)**. Presentations will take place during Week 11 & 12 in your Applied session. Attendance for both weeks is **mandatory**.

DVP Part 2: Report and Code

1. **Prepare a PDF report** (max 15 pages) and a **ZIP file** containing the *source code* for your narrative visualisation and *any data files* that are required to run it.
2. Name the files using the following format:
 - a. *StudentName_StudentID_Report.pdf*
 - b. *StudentName_StudentID_Code.zip*
3. Submit both files via Moodle under *Assessments* → *DVP Part 2: Visualisation Project Submission (37%)*. These **must** be two separate files. **Do not put your PDF inside of the ZIP archive**. Note that only .zip is recommended, and you should **not** use other extensions such as .rar or .7z.

Submit a **PDF report** and a **zip file containing your code and the data** to Moodle, due **the first week of exam period (see Moodle for date and time)**.

Notes on submissions

A few items to keep in mind while submitting.

- **We cannot mark any work submitted via email or stored on file hosts such as Google Drive or Dropbox or Github.** Please ensure that you submit correctly via Moodle since it is only in this process that you complete the required student declaration, without which your work cannot be assessed.
- Your assignment **MUST** show a status of "Submitted for grading" before it will be marked. If your submission shows a status of "Draft (not submitted)", **it will not be assessed and will**

incur late penalties if submitted after the due date/time. Note that this applies even if your file was uploaded to Moodle as draft prior to the due date.

- It is your responsibility to **ENSURE** that the files you submit are the correct files. We strongly recommend after uploading a submission, and prior to actually submitting on Moodle, that you download the submission and double-check its contents.
- Moodle may indicate an error in generation of a Turnitin report for the .zip file. This is not an issue to be concerned about. (It is not possible for us to turn on the Turnitin report only for one of the submitted files)
- Turnitin reports may not be available until after the submission. They are used to help staff review the academic integrity for all submissions. It may not be shared with students unless a student's work is under review.
- Moodle only allows a maximum file size of 500MB for submissions. This is rarely hit by students in the unit, but it can cause an issue if your data files are very large. If you believe the limit affects you, check your zipped folder size and look to reduce the size of your data (e.g., by removing columns you are not using). If this is not possible, then **only then** can you consider storing your data remotely, e.g., via Google Drive, but be sure to test your code and provide access. Be sure to note this restriction in your code comments and any instructions, if needed. If access and instructions are not provided, your mark will be penalised.
- You **do not** need to publish your app on the web.

Late Submissions and Special Consideration

DVP Part 1: Design Presentation

- We encourage everyone to submit their presentation slides on time.
- All students are required to attend Applied sessions in Week 11 and Week 12.
- **Design Presentation Slides submitted late will receive zero marks**, with no late penalty indicated on Moodle. Exemptions only considered if **Special Consideration** is granted. Please refer to the relevant section on the Assessment page on Moodle.

DVP Part 2: Report and Code

- Assessments received after the submission deadline, or after the extended submission date for those with special consideration, will be **penalised 5% of the available total marks per day up to a maximum of seven days**. Submissions more than seven days after the due date will not be graded.
- For information on eligibility for **Extensions** and **Special Consideration**, please refer to the relevant section on the Assessment page on Moodle.